# AI Traffic Light Systems Research Report

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# Abstract

With a constantly growing population and increasing road utilization, traffic congestion is becoming an issue that impacts areas from small towns to metropolitan cities. With the main focus being on the larger centralized hubs of transport, smaller towns such as Gainesville, Florida still face this challenge. The city's traffic system, infrastructure, and commuter population create a unique environment of unpredictable traffic patterns. This study explores the viability of artificial intelligence (AI) in helping manage and mediate delays. By assessing the problems and viewpoints of local residents, innovative and accommodating solutions can be designed to implement this emerging technology into existing systems. Many primarily travel by car, experience many delays due to traffic lights, believe environmental risks are a side effect, and have an optimistic outlook on AI. Additionally, relationships between data points suggest a road operator-oriented solution will be most beneficial to the general Gainesville community. The findings reveal a need for a way to ease congestion levels in the college town. Despite current technological and monetary limitations, AI is viewed to be a top contender to manage traffic lights in Gainesville.

# 1. Introduction

Within just over a century of industrialization, automobiles have become the most common form of transportation. As a result, systems have been implemented to enhance the safety and efficiency of traffic flows. A few modern systems are traffic lights, crosswalks, and vehicle-to-infrastructure communication. However, these technologies have not seen much change since their adoption. Congestion is a growing issue worldwide that creates delays, decreases productivity, and threatens air quality. With cities' continued rapid expansion to accommodate growing populations, more advanced solutions are being explored to mitigate these threats. Recent research supports the use of artificial intelligence in major metropolitan areas because of the many practical uses it already has. For example, fuzzy logic is being presented as a viable technique because of its versatile and accurate responses. Through this, traffic light and crosswalk timings can accommodate varying traffic levels [1]. However, similar problems are even prevalent in smaller towns, such as Gainesville, Florida.

Gainesville faces these challenges because of its inconsistently and unpredictably busy college town streets. With the University of Florida and its surrounding areas being main hubs for students, parents, sports fans, and tourists, using fixed traffic systems seems inefficient for these kinds of waves. This report will explore the proposal, framework, and implications of AI (artificial intelligence) in traffic management in Gainesville through research and data collecting.

# 2. Methods

The first step in finding if artificial intelligence is appropriate for helping to solve the traffic congestion issues in Gainesville is assessing residents' opinions on the state of traffic and AI.

Identifying the source and severity of the problem allows the development process to adapt to those conditions. The easiest and most accurate way was to survey a variety of local people who may or may not experience delays. From then, a target for the solution can be aimed at to approach with accuracy and relevancy.

## 2.1. Population/Sample

Collecting relevant data requires reaching an audience that is familiar with the topic being researched. For this circumstance, the ideal sample to gather and study was Gainesville residents who commute in the area often by car, bike, bus, walk, or any other mode. This population ensures the specificities of the topic at hand including localization and appropriate experiences. It also accounts for bias because one type of transportation may face more problems than another. The types of questions asked would give full freedom to the respondent to express their opinions on traffic light timings and attitudes toward artificial intelligence.

## 2.2. Materials/Instruments

Once core questions were identified to fit around human-centered design, a digital multiple-choice and scalable-response survey was designed in Qualtrics and distributed among the group's peers and community. By performing surveying with this tool, it was easily creatable, sharable, and analyzable. With many options for formatting, the prompts were made clear and concise. Additionally, results could be exported in many different ways to gather quantitative and qualitative data. After exportation, Microsoft Excel was utilized to better help in formulations and organization to better visualize responses and their relationships.

### 2.3. Procedure

Following the survey's fabrication in Qualtrics, the procedure involved two other steps: sharing and evaluating. First, the link to the survey was shared digitally with peers in the Gainesville area to complete and pass on to others in similar localities. Moreover, the link was shared in Canvas through discussion four. The goal of obtaining fifty or more respondents could be achieved through these methods so data can be dispersed and therefore more representative of the greater Gainesville. The second step was to take a deeper look at how people responded. The most important of the information dealt with opinions on light timings, which were exported from Qualtrics and organized in Excel to calculate averages and form charts. The same was done with other questions of the same answer type. Then to draw deeper conclusions, patterns were identified by relating the mode of transportation to the quality of standpoints. In the next section, the results from the gathering will be displayed to demonstrate the problem's acuteness and interrelatedness with the average Gainesville resident.

## 3. Results

### 3.1. Proportions

The data gathered came from over thirty individuals with varying transportation modes and opinions on traffic congestion in Gainesville. Many thoughtful insights on the topic were shared and can help gain a deeper understanding of the specific problems faced in this college town. Firstly, it was revealed that the majority of commuters travel by car, followed by on foot and other options as shown in Fig. 1. Those who selected "Other" specified that they use either a moped or e-scooter to get around.

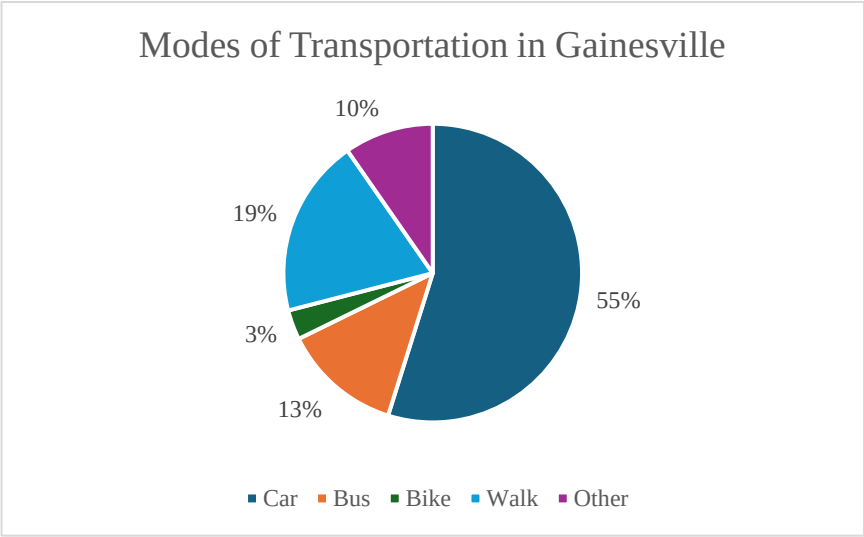


Fig. 1. Distribution of Commuting Methods

## 3.2. Traffic Light Timings and AI Perspectives

Secondly, the most profound results from the questions asked dealt with the timings of lights and perspectives on artificial intelligence's involvement in managing them. In Figs. 2 and 3, it shows the counts of what each answer choice got. Related, Figs. 4 and 5 express the feelings about AI in the context of the suggested solution to the problem.

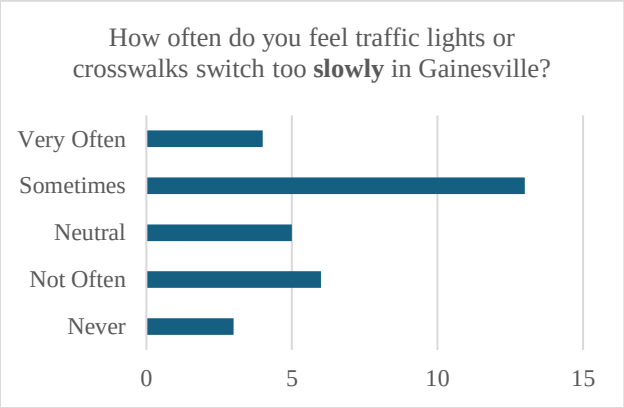


Fig. 2. Results of Question 2 with an average answer value of 3.29.

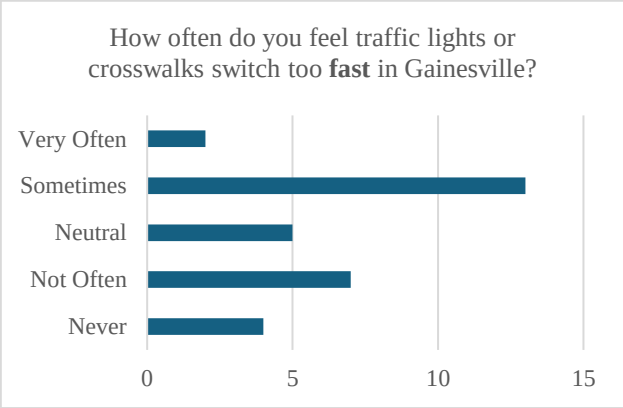


Fig. 3. Results of Question 3 with an average answer value of 3.06.

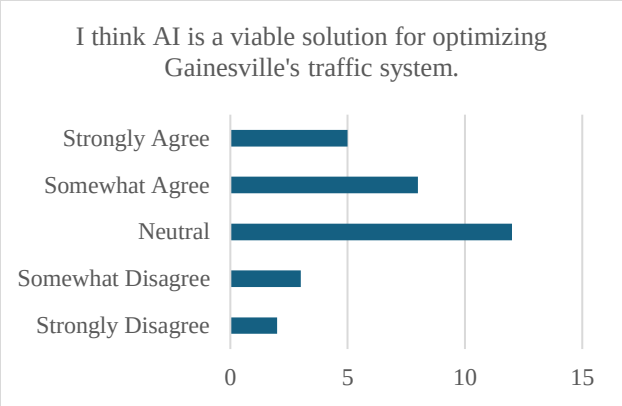


Fig. 4. Results of Question 7 with an average answer value of 3.48.

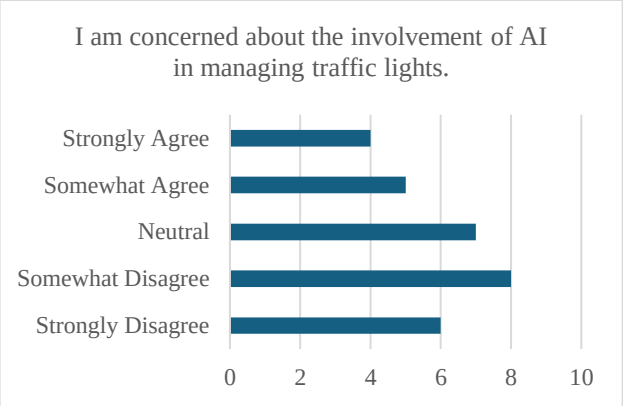


Fig. 5. Results of Question 8 with an average answer value of 2.86.

## 3.3. Relationships Between Datapoints

Lastly, the subcategorized results based on modes of transport show how the different ones show a relationship in how they navigate Gainesville. The proportionality in Fig. 6 indicates the operators on the road alter their routes for travel efficiency.

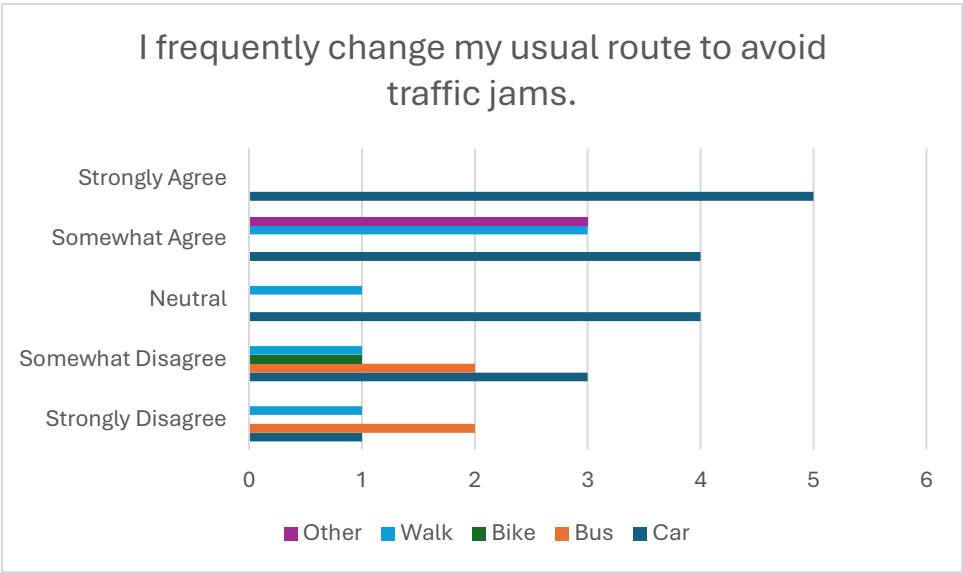


Fig. 6. Results of Question 5 including respondents' commuting preferences.

# 4. Discussion

This study's mission was to fully comprehend the traffic situation in Gainesville, Florida, and the possibility of implementing emerging technologies into the infrastructure by gathering opinions on light timings and artificial intelligence. The findings presented above demonstrate the variation of concerns among commuters and their relationships with efficiency.

## 4.1. Analyzation and Relations

The founding principle of the results is the proportion of those who most frequently travel by car and mopeds as opposed to on foot, bike, or bus. With traffic light timings mostly pertaining to drivers, this population knows and points out the issues in the current system. Reliance on this mode of transport suggests improvements in the management of traffic flows would benefit the majority of Gainesville. This is also an interesting statistic because, in contrast to another study conducted at Iowa State University, their population of car commuters makes up about a third of the total commuter population [2]. This highlights the uniqueness of Gainesville because it microscopically pinpoints aspects of the problem at hand. Many solutions for this college town must be curated for those behind the wheel.

Not only does the proportion support a proposed AI traffic management solution, but also compared to the other populations, they have overwhelmingly stronger feelings about the timing of lights. They say the lights switch inefficiently in that there are clear opportunities to go when the light is red. Additionally, nearly all of those who agreed with Question 4 were neutral or supportive of AI being a viable solution for Gainesville's traffic system. This is directly to

research that finds attitudes toward artificial intelligence in assisting public services are generally positive. The article outlines how Norway citizens feel about computers and related systems aiding their everyday lives. Being knowledgeable of the work involved and having high trust in such systems are characteristics that make people feel comfortable with this idea [3]. Correspondingly, most people surveyed who agreed that AI is a viable solution are not concerned about the involvement of it on the streets.

Finally, the last important aspect to take away from the survey results is the analysis of the subcategorized data by commuting method. Based on Fig. 6 above, a correlation is noticed between the type of transportation and strategies to avoid congestion. It was found that car and moped users tend to adjust routes much more often than any other groups. Due to most delays being caused by traffic bottlenecks, the impact of this affects these commuters in particular. A cause is poorly timed traffic lights in high vehicle volume areas as they stop traffic, even when it can be more efficient to continue the flow [4]. By building upon the existing system using AI, traffic levels can be accounted for in various circumstances using fuzzy logic and sensing technologies to ease the wait times for everybody on the road.

## 4.2. Reflection

The results and their interpretations emphasize a need for additional measures to the growing traffic issues in Gainesville, Florida. According to the analysis, artificial intelligence has become a clear and widely accepted answer among commuters. Car, bus, and moped users can benefit from reduced unnecessary stops, and pedestrians and bikers with similar routes can mutually benefit. Given the sample types, the research is specific to Gainesville's conditions, and therefore

may not apply to all environments. Nonetheless, principles can be useful to implementations in the broader scope.

Because artificial intelligence is an emerging technology, there are a few drawbacks to the proposal. The speed and cost of constant interpretation and transmission of live video feeds hinder the practicality and sustainability of this management method. Thorough, clear, and secure data must be communicated between components of the system to safely and efficiently reduce traffic delays [5]. While such limitations may exist now, continued development and enhancements can bring towns like Gainesville closer to a commuting experience with minimal delays.

Overall, this study suggests an approach involving AI can help heal the state of travel times through timely traffic lights and crosswalks in Gainesville, Florida. By striking a balance between efficiency and safety, it has the potential to positively impact the town's fast-paced and productive atmosphere.

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